HCG-50盐氯机酸碱平衡风险控制风险说明

Risk Statement for Acid Base Balance Risk Control of HCG-50 Salt Chlorine Machine

袁帅

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1、HASA原型机酸碱平衡控制说明；

HASA Prototype Acid Base Balance Control Instructions;

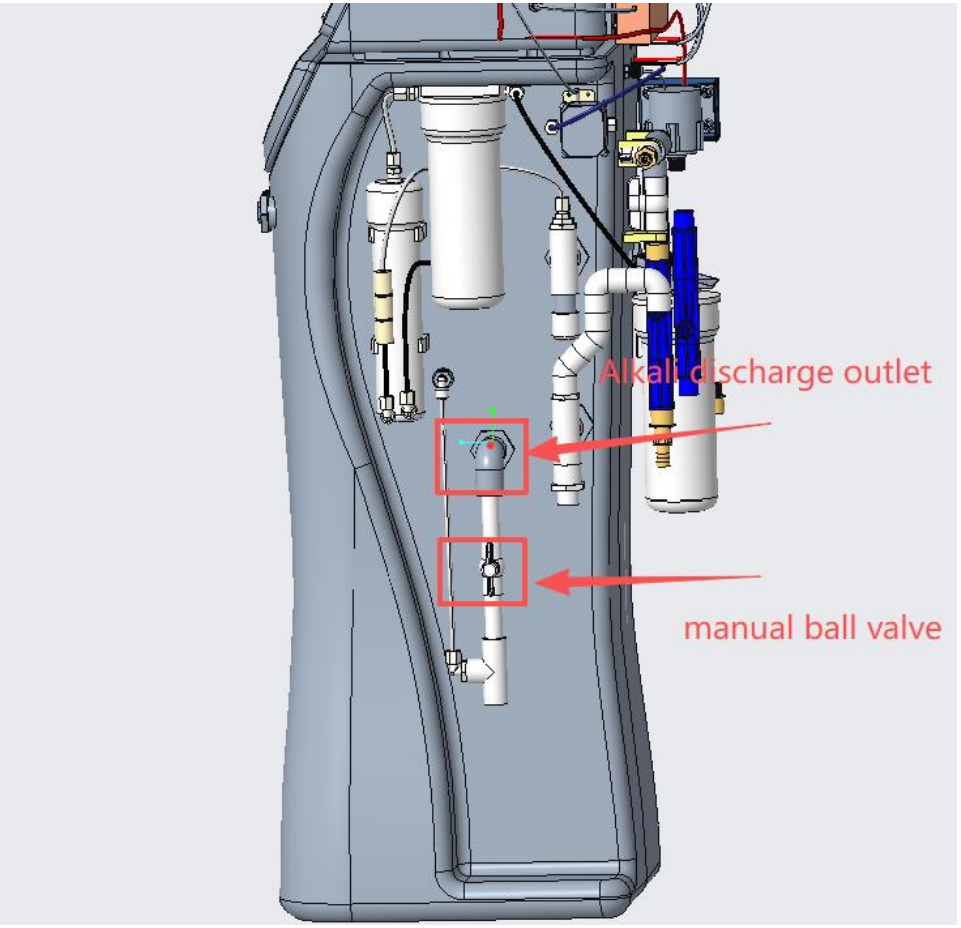
2、DVT样机酸碱平衡控制说明；

DVT Prototype Acid Base Balance Control Instructions;

3、排碱方案设想与挑战；

Assumption and challenges of alkali discharge plan;

1、HASA原型机酸碱平衡控制说明HASA Prototype Acid Base Balance Control Instructions;

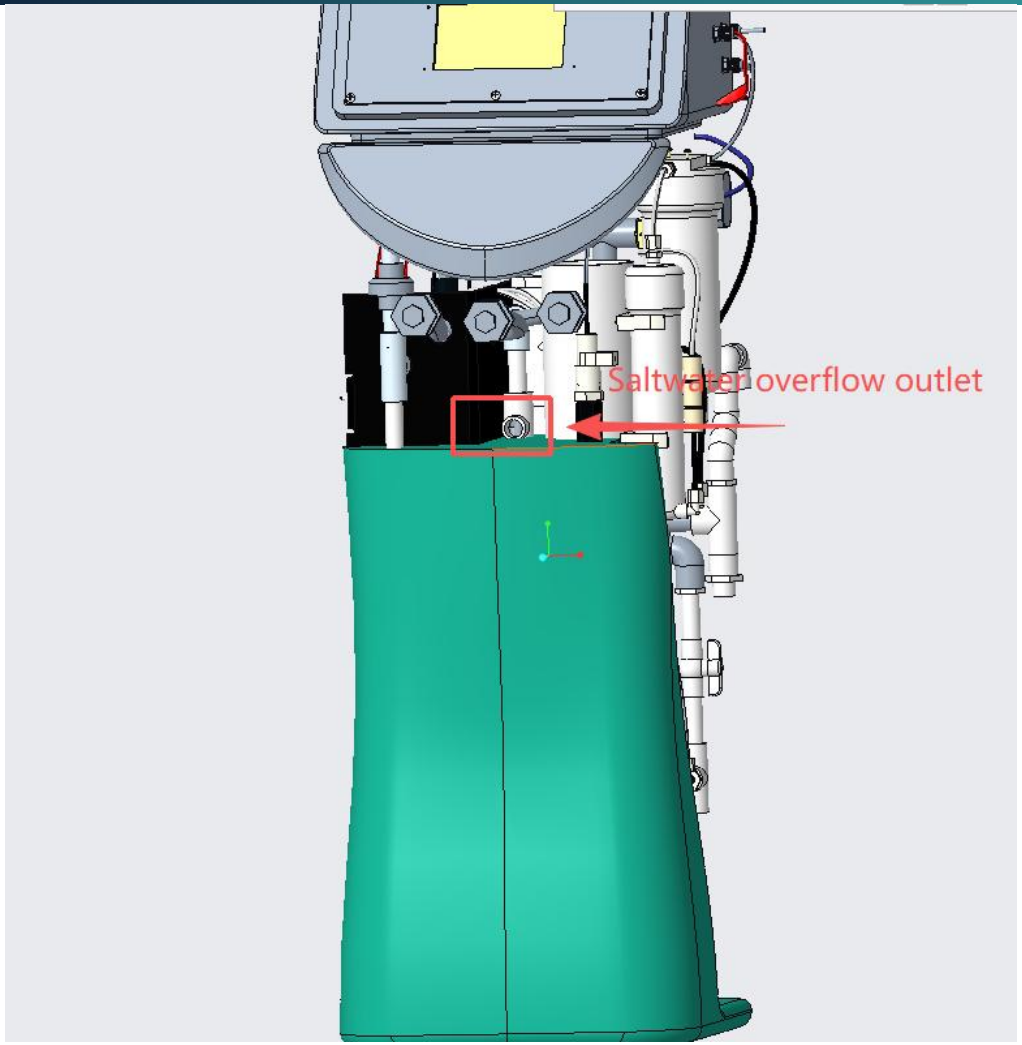


1、在HASA的原型机上面，碱液的排放需要手动操作，碱液排放口距盐水箱底部高度为57.2cm，由于碱水箱和盐水箱在49.5cm高度处相连接，这代表当打开碱液排放阀可以使盐水箱和碱水箱的液位一起下降到57.2cm的高度；

On the prototype of HASA, the discharge of alkali solution needs to be manually operated. The height of the alkali solution discharge port from the bottom of the salt water tank is 57.2cm. As the alkali water tank and the salt water tank are connected at a height of 49.5cm, this means that when the alkali solution discharge valve is opened, the liquid levels of both the salt water tank and the alkali water tank can be lowered to a height of 57.2cm;

2、原型机操作说明的描述，氯气精灵会产生氢氧化钠，用于帮助管理泳池的 pH 值。多余的氢氧化钠需要通过 pH 排放阀从氯气精灵中排出，以管理泳池的 pH 值。氢氧化钠是一种碱性溶液，需要从设备、其他泳池设备、硬质景观、植物和房屋通风口排出。

Description of prototype operation instructions,The Chlorine Genie produces sodium hydroxide which is used to help manage the pH in the pool. Excess sodium hydroxide needs to be drained from the Chlorine Genie through the pH Discharge Valve to manage the pH in the pool. The sodium hydroxide is an alkaline solution that needs to be discharged away from the unit, other pool equipment, hardscape, plants and house vents.



3、在HASA原型机的资料描述中，关于碱液排放的作用更多是为了调节泳池的PH，如果需要提高泳池PH就打开球阀，使碱液排放到泳池，如果不需要就保持这个球阀的关闭；如果泳池 pH 值过低（低于 7.2 ppm）：应关闭 pH 阀；如果泳池 pH 值过高（高于 7.8 ppm）：应打开 pH 阀；

In the description of the HASA prototype, the function of alkali discharge is mainly to regulate the pH of the swimming pool. If it is necessary to increase the pH of the swimming pool, open the ball valve to discharge the alkali into the pool. If it is not necessary, keep the ball valve closed;If pool pH is too low (below 7.2 ppm): pH Valve should be closed•； If pool pH is too high (above 7.8 ppm): pH Valve should be open；

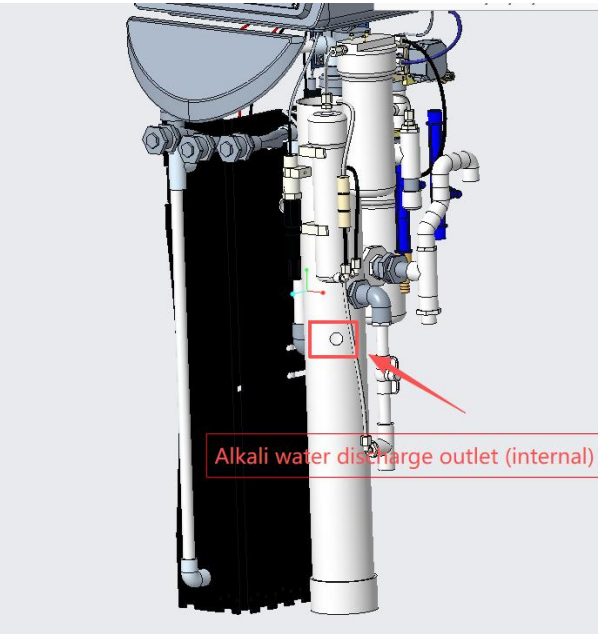
4、盐水箱的正常液位高度由氯气混合装置的溢流口决定，该高度距盐水箱底部为62cm,这代表每次排掉碱液的高度为4.8cm(62cm-57.2cm)；

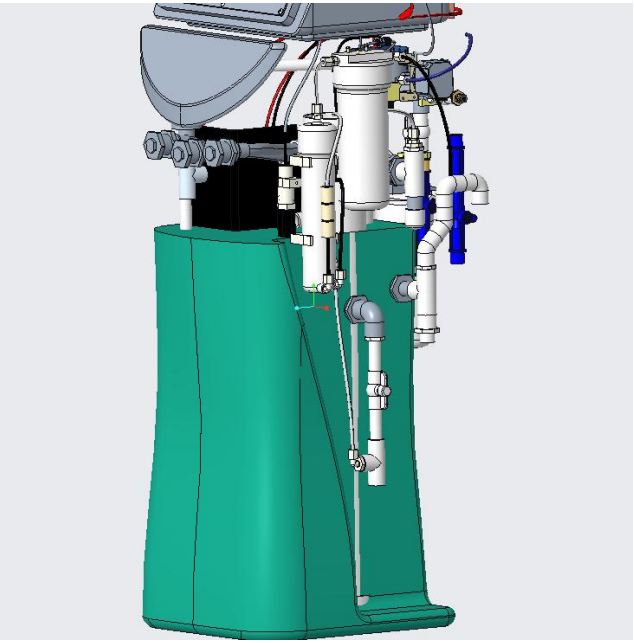
The normal liquid level height of the salt water tank is determined by the overflow port of the chlorine gas mixing device, which is 62cm from the bottom of the salt water tank. This means that the height at which alkali is discharged each time is 4.8cm (62cm-57.2cm);

1、HASA原型机酸碱平衡控制说明HASA Prototype Acid Base Balance Control Instructions;

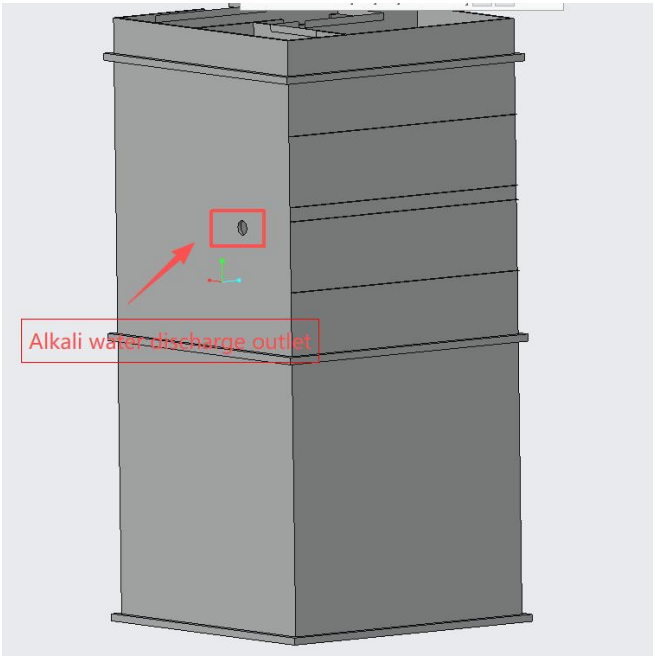
在HASA原型机上的酸碱平衡调节依赖外部监测数据支撑，手动调节也存在很大误差，只能做一个粗略的控制，但是有一些疑问例如碱水箱与盐水箱的联通为什么需要在盐水的液位之下，这样盐水箱和碱水箱始终联通而且液位高度一直，排放碱液的同时也会排放出盐水；

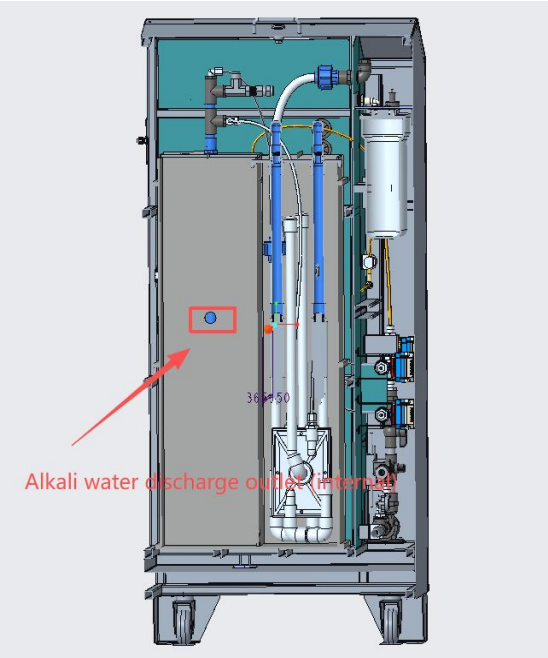
The acid-base balance adjustment on the HASA prototype relies on external monitoring data support, and manual adjustment also has significant errors, which can only be roughly controlled. However, there are some issues, such as why the connection between the alkali water tank and the salt water tank needs to be lower than the level of the salt water, so that the salt water tank and the alkali water tank are always connected and the level height of the two is always the same. When discharging alkaline solution, saltwater will also be discharged;

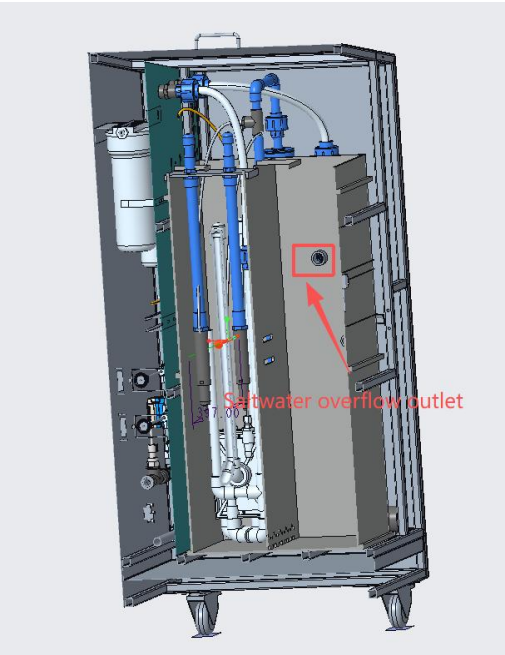




2、DVT样机酸碱平衡控制说明DVT Prototype Acid Base Balance Control Instructions;







1、在DVT样机的设计中，盐水溢流口高度（62cm)，碱水排放口高度（57.2cm），碱水排放口（内部）高度（49.5cm)，因为我们担心破坏这三个高度之间的关系可能会破坏整套系统的运行平衡，所以这三个高度与HASA原型机设计高度保持一致；

In the design of the DVT prototype, the height of the saltwater overflow port (62cm), the height of the alkaline water discharge port (57.2cm), and the height of the alkaline water discharge port (internal) (49.5cm) are kept consistent with the design height of the HASA prototype because we are concerned that disrupting the relationship between these three heights may disrupt the overall system balance;

2、DVT样机酸碱平衡控制说明DVT Prototype Acid Base Balance Control Instructions;

2、在DVT的设计中，碱液的排放由之前的无压排放更换为文丘里管驱动的负压排放，配合耐腐蚀的电动球阀D使用,可以做到智能远程控制排放；

In the design of DVT, the discharge of alkaline solution has been replaced by negative pressure discharge driven by a Venturi tube instead of the previous non pressure discharge. When used in conjunction with a corrosion-resistant electric ball valve D, intelligent remote control of discharge can be achieved；

3、这存在一些担忧，例如a、排放碱水会带出一部分盐水，这可能会提高泳池水中盐的含量；b、每次排放碱液的量是有限的（大约为4.8cm)，这对盐水PH的调节要求更加频繁,而碱液的产生量和浓度是一个变量，对排碱的控制有很大挑战；There are some concerns about this, such as a. Discharging alkaline water may bring out some saltwater, which may increase the salt content in the pool water; b、 The amount of alkali solution discharged each time is limited (approximately 4.8cm), which requires more frequent adjustment of the pH of the saline water. The amount and concentration of alkali solution produced are variables, posing great challenges to controlling the discharge of alkali solution;

4、我们有一些未来设计的猜想，是否可以提高碱水排放口（内部)的高度，使它高于盐水溢流口的高度，这样保持盐水和碱水分开，当然当碱水过多的时候依然会流向盐水，另外是否可以降低外部的碱水排放口，提高每次排放碱水的量，这样可能会降低碱水的排放频率；

We have some future design speculations. Can we increase the height of the alkali water discharge outlet (internal) to be higher than the height of the salt water overflow outlet, so as to keep the salt water and alkali water separated? Of course, when there is too much alkali water, it will still flow into the salt water. In addition, can we reduce the external alkali water discharge outlet and increase the amount of alkali water discharged each time, which may reduce the frequency of alkali water discharge;

3、排碱方案设想与挑战Assumption and challenges of alkali discharge plan;

1、排碱方案A:电解电极运行固定时间触发排碱机制（例如计数统计运行50分钟，排碱10分钟），停止电解--保持增压水泵运行--关闭电动球阀A/B/C---打开电动球阀D,液位传感器辅助判断排碱是否成功；

Alkali discharge plan A: The electrolysis electrode runs for a fixed time to trigger the alkali discharge mechanism (such as counting and counting for 50 minutes, alkali discharge for 10 minutes), stop electrolysis - keep the booster water pump running - close the electric ball valves A/B/C - open the electric ball valve D, and the liquid level sensor assists in determining whether the alkali discharge is successful;

2、排碱方案B:通过盐水PH触发排碱机制，PH过高--正常触发过程排放（停止电解--保持增压水泵运行--关闭电动球阀A/B/C---打开电动球阀D）---液位B脱离触发---液位C脱离触发--排碱结束--保持正常运行；

Alkali discharge planB:By triggering the alkaline discharge mechanism through salt water pH, if the pH is too high, the normal discharge process will be triggered (stop electrolysis - keep the booster water pump running - close electric ball valves A/B/C - open electric ball valve D) - liquid level B will be triggered - liquid level C will be triggered - alkaline discharge will end - maintain normal operation;

3、碱液浓度不固定，文丘里管驱动的负压排放受外部环境影响排放效率，电极监测数据有一定的滞后性，这些对碱液排放和酸碱平衡的控制有很大的挑战；

The concentration of alkali solution is not fixed, and the negative pressure discharge driven by the Venturi tube is affected by the external environment, resulting in a certain lag in electrode monitoring data. These pose great challenges to the control of alkali solution discharge and acid-base balance;